

Machine Learning

Proximity Measures

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Similarity and dissimilarity

- Similarity
 - Numerical measure of how alike two data objects are
 - Is higher when objects are more alike
 - Often falls in the range $[0,1]$
- Dissimilarity
 - Numerical measure of how different are two data objects
 - Lower when objects are more alike
 - Minimum dissimilarity is often 0
 - Upper limit varies
- Proximity refers to a similarity or dissimilarity

Similarity and Dissimilarity by Attribute type

p and q are the values of an attribute for two data objects

IN GENERAL
 $s = 1 - d$

Attribute type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$	$s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$
Ordinal Values mapped to integers 0 to $V-1$	$d = \frac{ p-q }{V-1}$	$s = 1 - \frac{ p-q }{V-1}$
Interval or Ratio	$d = p - q $	$s = \frac{1}{1+d}$ or $s = 1 - \frac{d - \min(d)}{\max(d) - \min(d)}$

Euclidean distance – L_2

$$\text{dist} = \sqrt{\sum_{d=1}^D (p_d - q_d)^2}$$

- Where D is the number of dimensions (attributes) and p_d and q_d are, respectively, the d -th attributes (components) of data objects p and q
- Standardization/Rescaling is necessary if scales differ

Minkowski distance – L_r

$$\text{dist} = \left(\sum_{d=1}^D |p_d - q_d|^r \right)^{\frac{1}{r}}$$

- Where D is the number of dimensions (attributes) and p_d and q_d are, respectively, the d -th attributes (components) of data objects p and q
- Standardization/Rescaling is necessary if scales differ
- r is a *parameter* which is chosen depending on the data set and the application

Minkowski distance – Cases

$r = 1$ also named *city block*, *Manhattan*, L_1 norm

- it is the best way to discriminate between zero distance and *near zero* distance
- a ϵ change on any coordinate causes a ϵ change in the distance
- works better than euclidean in very high dimensional spaces

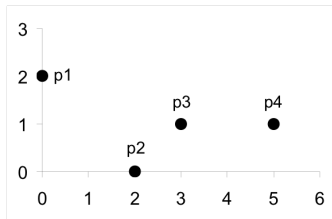
$r = 2$ euclidean, L_2 norm

$r = \infty$ also named Chebyshev, *supremum*, L_{max} norm, L_∞ norm

- considers only the dimension where the difference is maximum
- provides a simplified evaluation, disregarding the dimensions with lower differences

$$\text{dist}_\infty = \lim_{r \rightarrow \infty} \left(\sum_{d=1}^D |p_d - q_d|^r \right)^{\frac{1}{r}} = \max_d |p_d - q_d|$$

Minkowski distances – Example



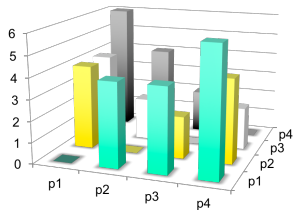
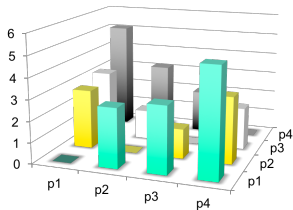
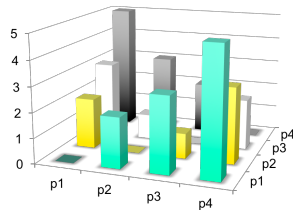
<i>point</i>	<i>x</i>	<i>y</i>
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L_1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L_2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_∞	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Comparison


 L_1

 L_2

 L_∞

Mahalanobis Distance

- Considers **data distribution**
- The Mahalanobis distance between two points p and q decreases if, keeping the same euclidean distance, the segment connecting the points is stretched along a direction of greater variation of data
- The distribution is described by the **covariance matrix** of the data set

$$\Sigma_{ij} = \frac{1}{N-1} \sum_{k=1}^N (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$\text{dist}_m = \sqrt{(p - q)\Sigma^{-1}(p - q)^T}$$

Mahalanobis Distance – Example

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

$$A = (0.5, 0.5)$$

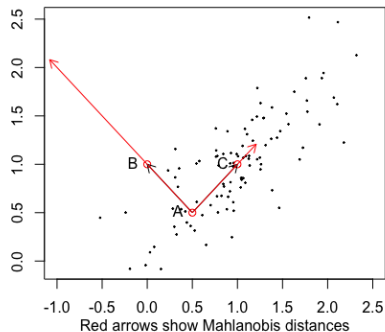
$$B = (0, 1)$$

$$C = (1, 1)$$

The euclidean distances AB and AC are the same

$$\text{dist}_m(A, B) = 2.236068$$

$$\text{dist}_m(A, C) = 1$$



Covariance matrix

- Variation of pairs of random variables
- The summation is over all the observations
- The main diagonal contains the variances
- The values are positive if the two variables grow together
- If the matrix is diagonal the variables are non-correlated
- If the variables are standardised the diagonal contains “one”
- If the variables are standardised and non correlated, the matrix is the identity and the Mahalanobis distance is the same as the euclidean

Common properties of a distance

1. **Positive definiteness:** $\text{Dist}(p, q) \geq 0 \ \forall p, q$
and $\text{Dist}(p, q) = 0$ if and only if $p = q$
2. **Symmetry:** $\text{Dist}(p, q) = \text{Dist}(q, p)$
3. **Triangle inequality:** $\text{Dist}(p, q) \leq \text{Dist}(p, r) + \text{Dist}(r, q) \forall p, q, r$

A distance function satisfying all the properties above is called a **metric**

Common properties of a Similarity

1. $\text{Sim}(p, q) = 1$ only if $p = q$
2. $\text{Sim}(p, q) = \text{Sim}(q, p)$

Similarity between binary vectors

- Consider the counts below

M_{00} the number of attributes where p is 0 and q is 0

M_{01} the number of attributes where p is 0 and q is 1

M_{10} the number of attributes where p is 1 and q is 0

M_{11} the number of attributes where p is 1 and q is 1

- Simple Matching Coefficient

$$\text{SMC} = \frac{\text{number of matches}}{\text{number of attributes}} = \frac{M_{00} + M_{11}}{M_{00} + M_{01} + M_{10} + M_{11}}$$

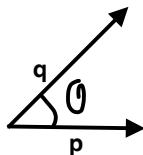
- Jaccard Coefficient

$$\text{JC} = \frac{\text{number of 11 matches}}{\text{number of non-both-zero attributes}} = \frac{M_{11}}{M_{01} + M_{10} + M_{11}}$$

Cosine similarity

- It is the cosine of the angle between two vectors

$$\cos(\theta) = \frac{p \cdot q}{\|p\| \|q\|}$$



- Example

$$p = 3 \ 2 \ 0 \ 5 \ 0 \ 0 \ 0 \ 2 \ 0 \ 0$$

$$q = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 2$$

$$p \cdot q = 3 * 1 + 2 * 0 + 0 * 0 + 5 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 2 * 1 + 0 * 0 + 0 * 2 = 5$$

$$\|p\| = \sqrt{3 * 3 + 2 * 2 + 0 * 0 + 5 * 5 + 0 * 0 + 0 * 0 + 0 * 0 + 2 * 2 + 0 * 0 + 0 * 0} = 6.481$$

$$\|q\| = \sqrt{1 * 1 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 1 * 1 + 0 * 0 + 2 * 2} = 2.245$$

$$\cos(p, q) = .3150$$

Extended Jaccard Coefficient (Tanimoto)

- Variation of Jaccard for continuous or count attributes
 - reduces to Jaccard for binary attributes

$$T(p, q) = \frac{p \cdot q}{\|p\|^2 + \|q\|^2 - p \cdot q}$$

Manhattan Distance – Use cases

- **Sparse High-Dimensional Data**

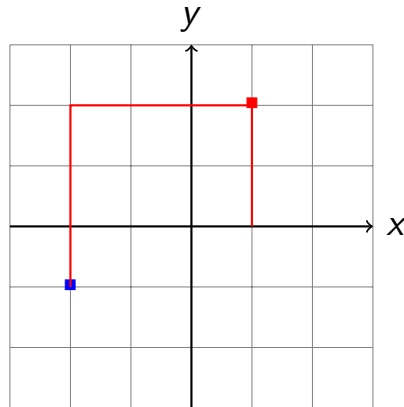
- Useful when features are not densely populated and dimensions are independent.

- **Grid-Based Systems**

- Applied in grid-like systems, such as in urban planning or robotics (e.g., pathfinding in grid maps).

- **Lasso Regression**

- Emphasizes feature selection by shrinking coefficients of less important features to zero.



Supremum (Chebyshev) Distance – Use cases

- **Anomaly Detection**

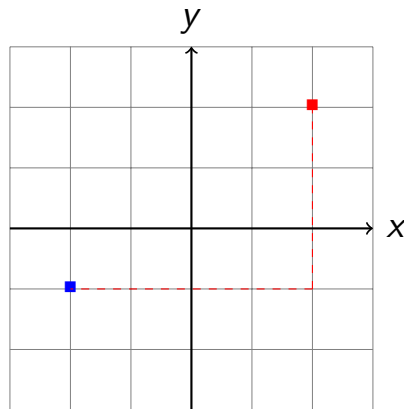
- Efficient for detecting extreme deviations across high-dimensional features.

- **Chessboard Distance**

- Often used in modeling real-world systems where maximum single-step moves matter (e.g., chessboard).

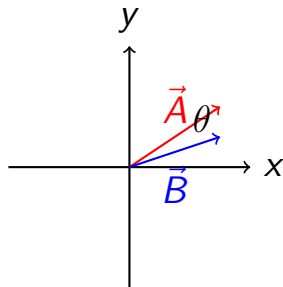
- **Industrial Applications**

- Applied in quality control, where maximum tolerances are checked for anomalies.



Cosine Similarity – Use cases

- **Text Mining and Document Similarity**
 - Used for document comparison and recommendation systems to detect contextually similar content.
- **Image Similarity**
 - Applied in image retrieval systems to match images with similar features.
- **Recommendation Systems**
 - Collaborative filtering methods often leverage cosine similarity to recommend items.



Jaccard Similarity

- **Definition:** Measures overlap between two sets relative to their union.
- **Use Cases**
 - **Document Similarity in Information Retrieval**
 - Applied in detecting plagiarism or duplicate documents.
 - **Image Processing**
 - Measures similarity in object recognition and segmentation.
 - **Clustering and Community Detection**
 - Useful in social network analysis to find communities based on shared elements.

